EFU ENERGY UTILITIES LICENCE v1.0

Experimental EFU-based Electricity, Heat, and Grid Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for energy utilities (electricity generation, district heating, grid operations).

NOT a power plant operating license, grid connection permit, or renewable subsidy replacement, but an open research protocol measuring energy production's physical footprint in human-scale EFU metrics (fuel, materials, land, heat waste). Covers coal, gas, nuclear, solar, wind, hydro, geothermal.

I. Core Concepts

1. **EFU-Energy (EFU-ENERGY)**
   EFU framework application for power generation and distribution:
   - **EFU-F_ENERGY**: fuel (coal, gas, uranium, biomass)
   - **EFU-M_ENERGY**: turbines, transformers, grid steel/aluminum
   - **EFU-L_ENERGY**: land use (plant footprint, transmission corridors)
   - **EFU-H_ENERGY**: waste heat (cooling water temperature rise)
   - **EFU-EM_ENERGY**: SOx, NOx, particulates, radionuclides
2. **Energy Sovereignty Gap (SS_ENERGY)**
   SS_ENERGY = TEFU_ENERGY_total – Regional Energy Carrying Capacity (RECC_ENERGY)
   Measures if energy production stays within regional resource regeneration + waste heat dissipation capacity.
3. **MROI_ENERGY**
   MROI_ENERGY = THI_ENERGY / TEFU_ENERGY
   THI_ENERGY = electricity supply reliability + decarbonization rate + grid resilience.

II. EFU Energy Licence – Research Status

Documents EFU-audited power plants, grid operators, district heating systems. **Does NOT replace** safety authorizations, emission permits, or fuel supply contracts. **Enables** coal vs. solar farm benchmarking on unified EFU scale.

III. Metrics and Thresholds

1. EFU Components (per TWh generated)

EFU-F_ENERGY (Fuel)
EFU-F_ENERGY = fuel_t / 300kg/person/year / TWh
Example: Coal = 300kt/TWh → 0.001 EFU/TWh

EFU-M_ENERGY (Materials)
EFU-M_ENERGY = steel_copper_t / 30kg/person/year / TWh

EFU-L_ENERGY (Land)
EFU-L_ENERGY = land_km² / 100km²/person/year / TWh

TEFU_ENERGY = 1.3×EFU-F + 1.5×EFU-M + 1.7×EFU-L + 1.4×EFU-H + 1.6×EFU-EM

2. Regional Energy Carrying Capacity (RECC_ENERGY)

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RECC_ENERGY = Sustainable_TWh / Population / 5MWh/person/year
Example: Germany = 500TWh / 84M / 5MWh = 0.12 EFU_ENERGY/person/year
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IV. Licence Categories (EFU-ENERGY Labels)

Level	SS_ENERGY	MROI_ENERGY	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-ENERGY SOVEREIGN**	Renewables + efficiency
2	0 < ≤30k EFU	≥5e-3	**EFU-ENERGY BALANCED**	Gas + growing renewables
3	30k < ≤150k EFU	≥1e-3	**EFU-ENERGY DEFICIT**	Coal/gas dominant
4	>150k EFU	<1e-3	**EFU-ENERGY CRITICAL**	Fossil lock-in

V. Regional RECC_ENERGY Calibrations

Country	Generation (TWh/year)	Population (M)	RECC_ENERGY (EFU/person/year)
GERMANY	500	84	0.12
USA	4,200	340	0.031
CHINA	8,500	1,410	0.004
FRANCE	500	68	0.12
EU	3,000	448	0.015

VI. Example: German Energy Sector (2025)

Input (annual sector):

- Generation: 500TWh
- Coal: 100TWh (30Mt coal)
- Gas: 150TWh (12B m³)
- Nuclear: 0TWh
- Renewables: 250TWh
- Grid losses: 40TWh
- Workers: 400,000 staff
- Region: Germany (RECC_ENERGY = 0.12 EFU/person/year)

Calculation:

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EFU-F_ENERGY = 30Mt coal/300/500 = 0.0002 EFU/TWh
EFU-M_ENERGY = grid materials/30/500 = 0.0001 EFU/TWh

TEFU_ENERGY_total ≈ 1,800,000 EFU/year
Allocated = 400k × 0.12 = 48,000 EFU/year
SS_ENERGY = 1.8M – 48k = +1,752,000 EFU/year (Level 4: CRITICAL)

Result: EFU-ENERGY CRITICAL
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Technology Comparison: Solar PV = 0.08 EFU/TWh (Level 2: BALANCED)

VII. Implementation Notes

1. **EU ETS integration**: TEFU_ENERGY for generation emission allocations
2. **Capacity markets**: EFU-M_ENERGY basis for grid expansion charges
3. **Heat pump tariffs**: EFU-H_ENERGY metric for waste heat recovery incentives
4. **Data sources**: Tier 1 = fuel receipts, meter data; Tier 2 = ENTSO-E statistics